

SAUMYA SHAH

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PROFESSIONAL SUMMARY

M.Tech. Robotics & Artificial Intelligence student with hands-on R&D experience in vehicle instrumentation, data acquisition, embedded systems, CAN communication, and prototype development. Experienced in sensor and CAN data analysis, AI/ML-based vehicle signal analysis, and real-time monitoring using Raspberry Pi, ESP32, Python, MATLAB, and LabVIEW. Strong background in system integration, testing & validation, feasibility studies, proof-of-concept development, and technical documentation.

TECHNICAL SKILLS

Instrumentation	Sensors & Transducers, Data Acquisition, Vehicle Telemetry, Signal Analysis, Real-Time Monitoring, Testing & Validation, Calibration
Embedded Systems	ESP32-S3, Raspberry Pi 5, Arduino, Embedded C/C++, Arduino Framework, Embedded Linux, GPIO, Serial Communication
Communication	CAN Bus, CAN Protocol Analysis, USB-CAN, Modbus TCP/IP, Ethernet/IP, TCP/IP
Programming & Tools	Python, C/C++, MATLAB, LabVIEW, Linux, Git, Data Visualization
Industrial Automation	PLC Programming, Mitsubishi PLC, Allen-Bradley PLC, HMI Development, Analog/Digital I/O
Engineering	System Integration, Prototype Development, Feasibility Studies, Proof-of-Concept Development, Hardware Testing, Troubleshooting, Technical Documentation

EXPERIENCE

Suzuki R&D Center India Pvt. Ltd. <i>Research & Development Intern – Robotics & AI</i>	May 2026 – Present <i>India</i>
- Developing embedded vehicle data acquisition and telemetry systems for the MITRA/VMEM micro e-mobility platform using Raspberry Pi 5, ESP32, and USB-CAN interfaces. - Performing reverse CAN engineering and sensor-data analysis to identify and correlate vehicle signals including battery current, motor current, wheel speed, and steering-related data. - Applying AI/ML techniques to sensor and CAN data for pattern recognition, signal analysis, and data-driven vehicle system monitoring using Python and MATLAB. - Using MATLAB and LabVIEW for measurement, data analysis, visualization, and experimental evaluation of vehicle and sensor data. - Supporting feasibility studies and proof-of-concept development for low-cost vehicle instrumentation, data acquisition, telemetry, and monitoring systems, followed by controlled testing and validation. - Performing hardware and system-level testing, troubleshooting communication, wiring, power, and data-acquisition issues, and preparing technical documentation covering experimental methodology, CAN signal mapping, and validation results.	
TatvaSoft <i>Software Developer Trainee</i>	May 2024 – Jun. 2024 <i>India</i>
Worked on software development tasks involving application development, debugging, database operations, and implementation of software features.	

SELECTED PROJECTS

MITRA (VMEM) – CAN Bus Reverse Engineering & Vehicle Telemetry	<i>Raspberry Pi 5 CAN Sensors DAQ</i>
- Reverse engineered CAN communication on the MITRA/VMEM platform to investigate battery and motor current-related vehicle signals. - Logged CAN traffic and independently measured battery-current data using Raspberry Pi 5, USB-CAN, WCS1800 Hall-effect current sensor, and ADS1115 ADC. - Correlated time-aligned sensor measurements with CAN IDs and byte fields to identify candidate vehicle-current signals and validate signal behavior. - Defined controlled validation conditions covering stationary operation, straight-line motion, turns, acceleration, deceleration, and varying loads.	
Low-Cost Vehicle Data Logger	<i>ESP32-S3 Embedded C/C++ Arduino Framework</i>
- Designed and developed an ESP32-S3-based low-cost vehicle data logger for vehicle data acquisition and monitoring. - Developed Embedded C/C++ firmware using the Arduino framework for sensor interfacing, data acquisition, and real-time	

logging.

- Integrated vehicle sensing hardware with the ESP32-S3 and verified signal acquisition, power regulation, and hardware interfaces.
- Performed component and system-level testing to validate wiring, power supply, sensor connectivity, and logger operation.

PLC-HMI Industrial Automation & Control System

Mitsubishi | Allen-Bradley | Modbus TCP/IP

- Developed ladder-logic-based control applications using Mitsubishi and Allen-Bradley PLCs with industrial communication interfaces.
- Configured Modbus TCP/IP and Ethernet/IP communication for PLC-HMI system integration and real-time monitoring.
- Designed HMI dashboards for system status, alarms, monitoring, and visualization of process parameters.
- Performed analog/digital I/O validation, system debugging, and communication troubleshooting.

Vision-Guided Robotic Sorting System

UR10e | Industrial Camera | Computer Vision

- Developed a vision-guided robotic sorting system integrating an industrial camera, UR10e robot, and real-time control software.
- Performed camera-to-robot calibration and coordinate transformation for accurate object localization and robotic manipulation.
- Integrated perception, calibration, communication, motion planning, and robot control into a complete prototype system.

Smart Energy Monitoring System

Embedded Systems | Sensors | IoT

- Designed an appliance-level energy monitoring architecture using embedded sensing for electrical parameter measurement and data acquisition.
- Integrated sensors and connected monitoring interfaces for real-time energy visualization and analysis.

EDUCATION

M.Tech. in Robotics & Artificial Intelligence

Department of Electronics & Instrumentation Engineering, Nirma University

2025 – 2027

CGPA: 8.35

B.E. in Computer Engineering

LDRP Institute of Technology and Research

2021 – 2025

CGPA: 6.77

CERTIFICATIONS

- Python for Data Science
- Certificate of Participation – Tata
- Certificate of Participation – HP
- Certificate of Participation – Adobe

ADDITIONAL INFORMATION

Languages: English, Hindi, Gujarati

Interests: Embedded Systems, Vehicle Instrumentation, Data Acquisition, Robotics, Prototyping, Automotive R&D